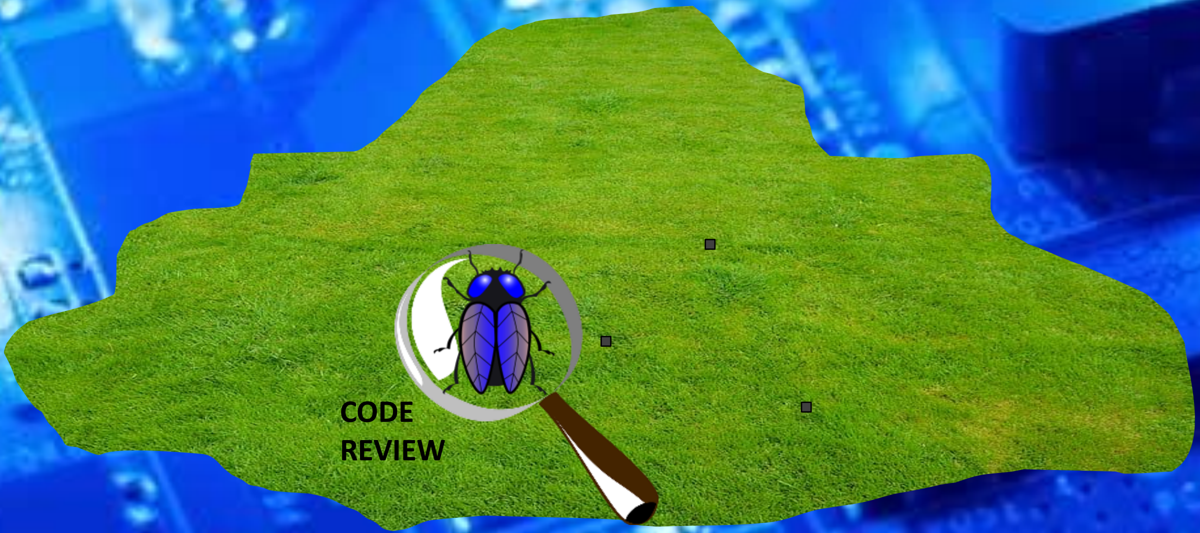
EEE3096S



Code reviews

Lecture L04X

Embedded Systems II

Dr Simon Winberg

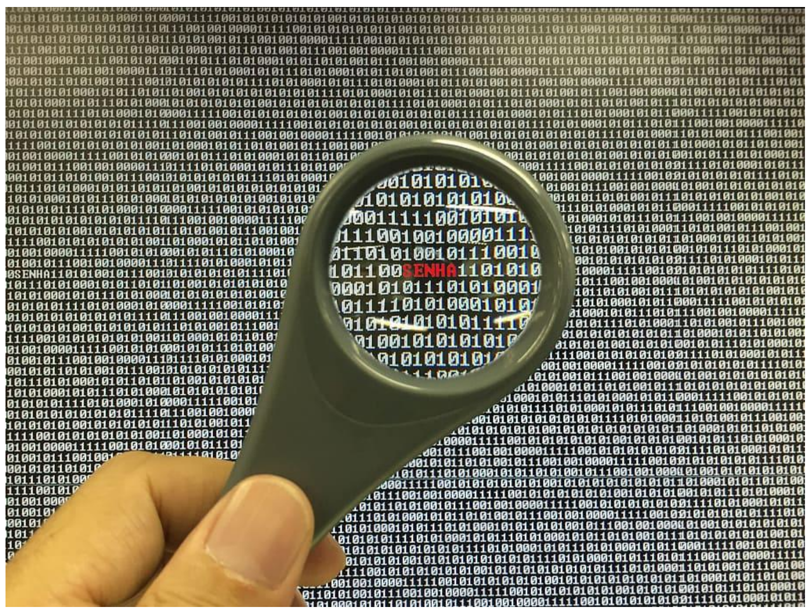
Electrical Engineering
University of Cape Town

Overview

- Code reviews

An ingredient of a professional engineer

The Code Review



Reason for a code review

- #1 reason for code reviews:
  To mitigate challenges of large code base worked on by multiple developers and to sustain quality*
- Focus is deciding how to ensure…
  - Maintainable code?
  - Readable code?
  - Bug-free code?
  - Exchangeable and portable code?

Do you know … what this is and how it's done?!

* as the code base of a project can be a signiant capital outline and asset to a company.

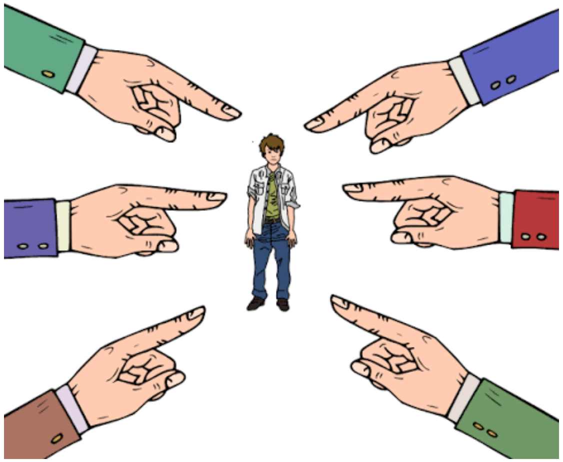
Reason for a code review

- It's also about managing quality of code and ways to assess this
  - Average defect detection rate in various testing
  - Unit testing
  - Function testing
  - Integration testing
  - How can this be improved?
- Can also be important aspect of getting products certified/approved and purchased.

The Code Review Practice

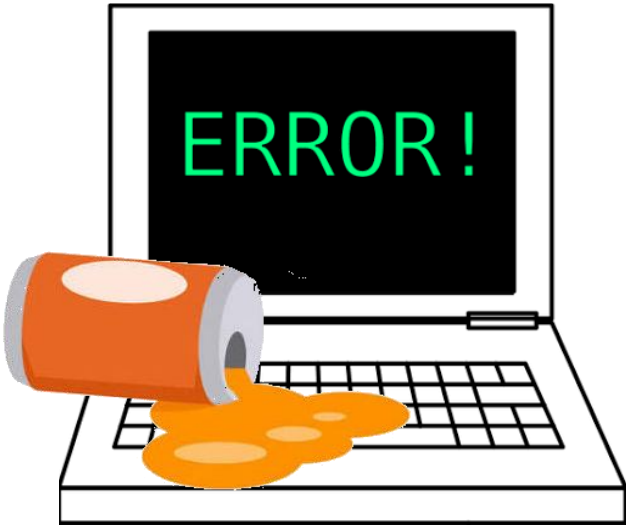
- It involves a constructive review of fellow developer(s) code.
- More formally (in business) may require sign-off from another team member before a developer is permitted to check in changes or new code to the organization’s repository.
- Analogy to doing a group project report:
  - Do all the group members understand the purpose and aim of the task?
  - Is it well structured?
  - Is the grammar good and the spelling checked?
  - Are all the pieces in place? Nothing missing?
  - Are the results and conclusions clear and correct?

The Who-What-When (wowawe)



Who

is responsible?



What

was found/needs review

When

did it happen

The Who-What-When of Code Reviews

- **WHO** was the original developer? Who is the reviewer? Both or either may be done separately or jointly.
- **WHAT** suggestions or feedback? The reviewer gives suggestions for improvement the algorithms, logical and/or structural level. Check the conformance to agreed styles and standards.

  notes: Feedback leads to refactoring (assigned possibly the the original author or other team member), followed by a 2nd code review (by the same of different team member). Eventually code gets approved, archives and the development process proceeds.
- **WHEN** is code checked, due dates for code approvals, etc. This is also a matter of deciding at what stage, a code review happens, e.g. once a certain amount of code (LOC) has been developed, or after a certain time period.

Why are code reviews important

- Ensures more than one person has seen every piece of code
- The potential of someone reviewing your code (and the peer pressure or possibly establishing an opinion of your expertise) help raise the quality.
- It forces the authors of the code to explain their decisions clearly

Why are code reviews important

- Helps newbie coders / new recruits, e.g.
  - It gives a useful learning experience for newbies, without degrading quality of the main code assets.
  - Helps in pairing new employees with appropriate experienced developers (e.g. based on code they're good at or need to develop)
- Encourages members of a large team to interact and know about other parts of the system.
- Reduces redundancy
- Enhances overall understanding
- Shares responsibility, e.g. code author and reviewer both accountable for committing code

Code Review Variants

- **Inspection:** A formalized code review with
  - defined roles usually assigned (e.g. author, moderator, reviewer, scribe … etc.)
  - several reviewers often look at the same piece of code to ensure thoroughness
  - checklist of flaws or other issues to look for
- **Walkthrough:** a less formal discussion of code between author and a single reviewer (or possibly a small number of reviewers)
- **Code Reading:** Reviewer(s) look at code by themselves, possibly without having a formal meeting, maybe just emailing suggstions.

Concluding points on code reviews

Code reviews are certainly used frequently in industry, typically in both smaller and certainly in the more massive companies (such as Google, Facebook, Microsoft, etc.)

Code reviews regarding this course… well, for most purposes it would be quite excessive (e.g. for pracs) but you may want to get into this sort of practice, e.g. you and your prac partner doing a quick code reading of each other's work prior to submitting.

End of Lecture